Weekly Review: VLAN and Inter-VLAN Routing

Cisco Packet Tracer Practical Report

1. Objective

Build a three-switch network with two VLANs (Sales and Support), trunk the switches together, and enable communication between the VLANs using router-on-a-stick.

2. Topology and Addressing

Devices: 1 router (2911), 3 switches (2960: SW1, SW2, SW3), 6 PCs. SW1 connects to the router and to SW2 and SW3. Each switch has one Sales PC and one Support PC.

```
            Router R1
               | Gi0/0
               | Gi0/1
    SW2 ----- SW1 ----- SW3
    (Fa0/1 to SW1 Fa0/1)   (Fa0/1 to SW1 Fa0/2)
```

PC	Switch / Port	VLAN	IP Address	Mask	Gateway
PC1	SW1 Fa0/3	10 Sales	192.168.10.11	255.255.255.0	192.168.10.1
PC2	SW1 Fa0/4	20 Support	192.168.20.11	255.255.255.0	192.168.20.1
PC3	SW2 Fa0/2	10 Sales	192.168.10.12	255.255.255.0	192.168.10.1
PC4	SW2 Fa0/3	20 Support	192.168.20.12	255.255.255.0	192.168.20.1
PC5	SW3 Fa0/2	10 Sales	192.168.10.13	255.255.255.0	192.168.10.1
PC6	SW3 Fa0/3	20 Support	192.168.20.13	255.255.255.0	192.168.20.1

Link	Port A	Port B
SW1 to R1 (trunk)	SW1 Gi0/1	R1 Gi0/0
SW1 to SW2 (trunk)	SW1 Fa0/1	SW2 Fa0/1
SW1 to SW3 (trunk)	SW1 Fa0/2	SW3 Fa0/1

3. Task 1: Create VLANs and Assign PC Ports

Create the VLANs on all three switches (shown for SW1; repeat on SW2 and SW3):

```
Switch> enable
Switch# configure terminal
```

```
Switch(config)# hostname SW1
SW1(config)# vlan 10
SW1(config-vlan)# name Sales
SW1(config-vlan)# vlan 20
SW1(config-vlan)# name Support
SW1(config-vlan)# exit
```

Assign the PC ports as access ports. SW1 (Fa0/3 Sales, Fa0/4 Support):

```
SW1(config)# interface fastEthernet 0/3
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# interface fastEthernet 0/4
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 20
SW1(config-if)# end
```

SW2 and SW3 (Fa0/2 Sales, Fa0/3 Support):

```
SW2(config)# interface fastEthernet 0/2
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access vlan 10
SW2(config-if)# interface fastEthernet 0/3
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access vlan 20
SW2(config-if)# end
```

Verify with show vlan brief on each switch.

4. Task 2: Configure and Verify Trunks

SW1 has three trunk ports (to SW2, SW3 and the router):

```
SW1(config)# interface range fastEthernet 0/1 - 2
SW1(config-if-range)# switchport mode trunk
SW1(config-if-range)# exit
SW1(config)# interface gigabitEthernet 0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# end
```

SW2 and SW3 each have one trunk port toward SW1:

```
SW2(config)# interface fastEthernet 0/1
SW2(config-if)# switchport mode trunk
SW2(config-if)# end
```

Verify with show interfaces trunk. Expected output on SW1:

```
SW1# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1        on           802.1q         trunking      1
Fa0/2        on           802.1q         trunking      1
Gig0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1        1-1005
Fa0/2        1-1005
Gig0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1        1,10,20
```

```
Fa0/2        1,10,20
Gig0/1       1,10,20
```

Screenshot to submit: the show interfaces trunk output from SW1 (and SW2/SW3 if possible) taken in Packet Tracer. Both VLAN 10 and 20 should appear in the allowed and active list.

Check: a Sales PC on one switch can ping a Sales PC on another switch (for example PC1 to PC3), which proves the trunks work. Support PCs likewise (PC2 to PC4).

5. Task 3: Router-on-a-Stick

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitEthernet 0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# exit
R1(config)# interface gigabitEthernet 0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# end
R1# copy running-config startup-config
```

Verify with show ip interface brief (Gi0/0, Gi0/0.10 and Gi0/0.20 up/up) and show ip route (two connected routes, 192.168.10.0/24 and 192.168.20.0/24).

Alternative: a multilayer switch can do the same with ip routing and an interface vlan 10 / interface vlan 20 SVI for each VLAN, with no router needed.

6. Task 4: Connectivity Tests and Troubleshooting

Test	Expected	Result (fill in)
PC1 to 192.168.10.1 (gateway)	Success	
PC1 to PC3 (Sales, different switch)	Success	
PC2 to PC4 (Support, different switch)	Success	
PC1 to PC2 (Sales to Support, same switch)	Success	
PC1 to PC6 (Sales to Support, different switch)	Success	
PC5 to PC4 (Sales to Support, different switch)	Success	

The first ping may time out while ARP completes; repeat it before deciding that it failed.

Troubleshooting guide

Problem	How to check	Fix
Wrong default gateway on a PC	ipconfig on the PC	Set the gateway to 192.168.10.1 or 192.168.20.1
Wrong sub-interface IP or mask	show ip interface brief on R1	Correct the ip address command
Wrong VLAN number in encapsulation	show running-config on R1	Use encapsulation dot1Q 10 / 20
Port to router is not a trunk	show interfaces trunk on SW1	switchport mode trunk on Gi0/1
VLAN not allowed or missing on a trunk	show interfaces trunk, show vlan brief	Create the VLAN; switchport trunk allowed vlan add 20
PC port in the wrong VLAN	show vlan brief	switchport access vlan 10 or 20
Router interface shut down	show ip interface brief	no shutdown on Gi0/0

Example fix recorded in this practical: pings from Sales to Support failed because the PC gateway was set wrongly. After correcting the gateway to the sub-interface IP, pings between VLAN 10 and VLAN 20 succeeded. (Replace with the fault you actually found.)

7. Conclusion

Three switches were connected with 802.1Q trunks, and PCs were assigned to VLAN 10 (Sales) and VLAN 20 (Support). Router-on-a-stick sub-interfaces provided inter-VLAN routing. Verification with show interfaces trunk and ping tests confirmed that VLAN traffic crosses the switches and that Sales and Support can communicate through the router.

8. Submission Checklist

- Topology screenshot with VLAN and IP labels
- show interfaces trunk screenshot
- show vlan brief and show ip route screenshots
- Ping results between VLAN 10 and VLAN 20
- Notes on any problem found and how it was fixed